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INTEGRATED SCIENCE GRADE 9

GRADE 9

FIRST EDITION

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INTEGRATED
SCIENCE

STRAND 1: MIXTURES, ELEMENTS S AND COMPOUNDS

Simple Structure of the Atom

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There are about 119 known elements

The following illustration shows the names and symbols of first 20 elements of the periodic table.

First Twenty Elements				
1 H Hydrogen	2 He Helium	3 Li Lithium	4 Be Beryllium	5 B Boron
6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon
11 Na Sodium	12 Mg Magnesium	13 Al Aluminum	14 Si Silicon	15 P Phosphorus
16 S Sulfur	17 Cl Chlorine	18 Ar Argon	19 K Potassium	20 Ca Calcium

Atoms are very small particles of an element, but they contain even smaller particles called sub-atomic particles as follows;

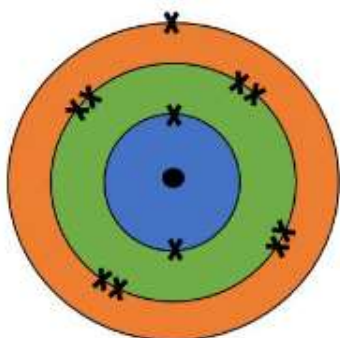
- **Protons** - found in the nucleus, they are positively charged (+ve)
- **Electrons** - found in the nucleus, they are negatively charged (-ve)
- **Neutrons** - they can be imagined as circulating the nucleus in energy levels, they have no charge
- In a neutral atom, *Number of protons = Number of electrons*

Particles present in an atom are summarized below.

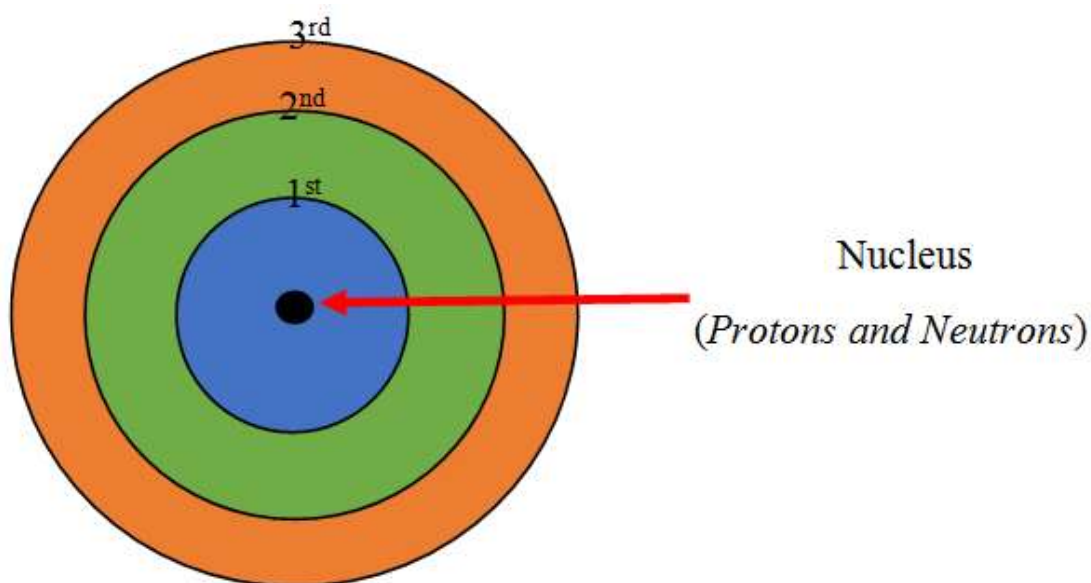
Sub-atomic Particle	Mass	Charge	Where found in atom
Proton	1	+ (Positive)	inside nucleus
Neutron	1	0 (Neutral)	inside nucleus
Electron	1/1840	- (Negative)	outside nucleus

Structure of the Atom

The various energy levels in an atom are represented by a series of circles sharing the same centre (nucleus), separated from each other by roughly equal distances



The nucleus of the atom is at the centre of the circles. The electrons in the energy levels are represented by dots (.) or crosses (x). The energy levels are labelled 1st, 2nd, 3rd, 4th and so on starting from the one nearest to the nucleus as shown below;



The electrons that occupy the 1st energy level have lower energy than those in the 2nd energy level. Subsequently those in the 2nd energy level have lower energy than those in the 3rd energy level and so on. The 1st energy level usually has a maximum of two electrons while the 2nd energy level has a maximum of eight (8) electrons.

We can summarise the electrons, and their arrangement in each energy level as shown in the table below.

Element	Symbol	Electron Configuration
Hydrogen	H	1
Helium	He	2
Lithium	Li	2.1

Element	Symbol	Electron Configuration
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Beryllium	Be	2.2
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Boron	B	2.3
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Carbon	C	2.4
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Nitrogen	N	2.5
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Oxygen	O	2.6
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Fluorine	F	2.7
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Neon	Ne	2.8
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Sodium	Na	2.8.1
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Magnesium	Mg	2.8.2
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Aluminium	Al	2.8.3
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Element	Symbol	Electron Configuration
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Silicon	Si	2.8.4
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Phosphorous	P	2.8.5
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Sulphur	S	2.8.6
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Chlorine	Cl	2.8.7
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Argon	Ar	2.8.8
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Potassium	K	2.8.8.1
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Calcium	Ca	2.8.8.2
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Atomic Characteristics

Atomic Number and Mass Number

The atomic number tells us *how many protons* are there in a nucleus. It is denoted by letter Z.

It also tells us the number of electrons in an atom
Atomic Number = Number of Protons = Number of Electrons

The mass number is the sum of protons and neutrons, therefore it is always bigger than the atomic number. It is denoted by letter A. Roughly the mass number is double the atomic number.

To get the number of neutrons, we just subtract the atomic number from the mass number, that is $A - Z$.

Atom	Symbol	Number of Protons: Atomic number, Z	Neutrons	Mass Number, A
Hydrogen	H	1	0	1
Carbon	C	6	6	12
Nitrogen	N	7	7	14
Sodium	Na	11	12	23
Chlorine	Cl	17	18	35

Example

Calculate the number of neutrons in a chlorine atom given that the atomic number, Z = 17 and mass number, A = 35

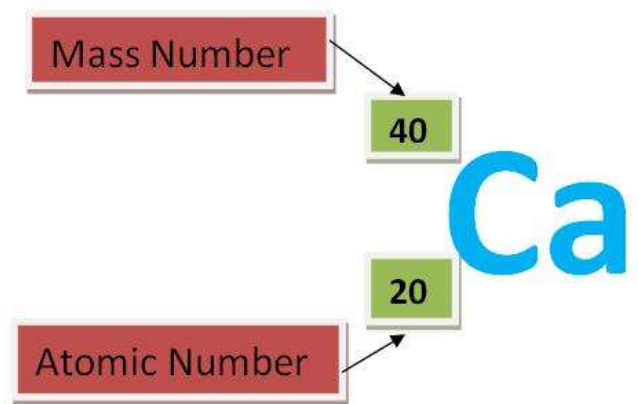
The number of neutrons

$$= A - Z$$

$$= 35 - 17$$

$$= 18$$

Usually, the atomic number, Z , and mass number A , of an atom of an element X can be written alongside the symbol of that element, one as a superscript and the other a subscript as shown below



NOTE: The top number is referred to as the superscript and bottom number as the subscript

Metal and alloys

Alloys

Metal alloys are created by combining two or more elements to form a new material. These alloys can be composed entirely of metals or a mix of metals and non-metals. While the combination results in a new alloy, it retains the properties of the original metals, such as electrical conductivity, opacity, ductility, and luster.

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However, they may exhibit traits distinct from pure metals, such as increased hardness or strength. In some cases, an alloy can retain the essential properties of a metal while reducing overall material costs. Additionally, the chemical combination in alloys can impart synergistic qualities to the component metals, such as enhanced corrosion resistance or improved mechanical strength.

For practical applications, alloy components are usually measured by mass percentage, while atomic fraction is used in fundamental scientific research. Alloys are categorized based on their atomic arrangement: substitutional alloys, where some atoms of the base metal are replaced by another element; interstitial alloys, where smaller atoms fit into the gaps between metal atoms; and heterogeneous alloys, which consist of two or more distinct phases. Additionally, alloys can be classified as intermetallic compounds or homogeneous alloys, which have a single phase. An alloy may either be a blend of different metallic phases or a solid solution, where all metal grains (crystals) share the same composition.



Characteristics of Alloys

- ✓ An alloy is a mixture of chemical elements that creates a material with metal-like properties, combining at least one metal with other elements. Unlike impure metals, such as wrought iron, which are less controlled but still useful, alloys are carefully engineered to achieve specific characteristics. The main metal in the alloy is often called the base metal or primary metal, and the alloy may be named after it.
- ✓ The additional elements in an alloy, which may or may not be metals, are soluble in the molten base metal and integrate into the mixture.
- ✓ The mechanical characteristics of alloys are frequently very different from those of their base metal. Alloying a metal with another soft metal, such as copper, can change a metal that is typically highly soft (malleable), such as aluminum.
- ✓ Despite the fact that both metals are relatively ductile and soft, the final alloy of aluminum will be stronger.
- ✓ By adding a small amount of non-metallic carbon to iron, its excellent ductility is exchanged for the increased strength of an alloy known as steel. Steel is a highly practical and widely-used alloy due to its remarkable strength, substantial toughness, and the ability to undergo significant modification through heat treatment.
- ✓ Additionally, steel can be enhanced for specific purposes: chromium can be added to improve corrosion resistance (resulting in stainless steel), or silicon can be incorporated to enhance electrical conductivity (producing silicon steel).

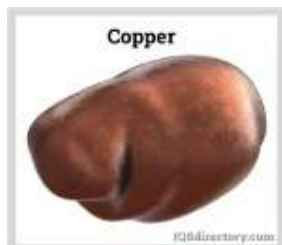
Metals

- Pure metals consist of a single type of atom and exhibit a uniform structure, making them single-phase metals. In contrast, alloyed metals are combinations of pure metals mixed in specific percentages to create a composite with distinct properties.
- Pure metals have defined mass, melting point, and physical characteristics. However, pure metals, with some exceptions, generally lack the strength, toughness, and durability of alloyed metals, which is why they are often used as components in alloys.
- All metal products incorporate a form of pure metal that has been alloyed. While the periodic table lists numerous pure metals, only a few are widely used in commercial products and are present in various metal items.
- Metals such as iron, aluminum, copper, zinc, titanium, chromium, and nickel are commonly used to manufacture both commercial and industrial products.
- Alloys derived from pure metals play a crucial role in modern society. Metallurgists and engineers continually explore innovative methods to blend pure metals, creating alloys that are both strong and durable. During the Bronze Age, early metallurgists developed techniques to mix pure metals, resulting in stronger and more resilient products. To qualify as "pure," a metal must consist of 99% or more of that metal without any alloying.

Copper

- Copper was among the first metals discovered by humans. Known for its remarkable malleability, it also boasts excellent thermal and electrical conductivity.
- Copper's resistance to corrosion and its durability make it a valuable material. Unlike many pure metals, copper is frequently utilized in its pure form, especially in electronic products, due to its superior conductivity.
- Additionally, its resilience in high-moisture environments contributes to its long-lasting nature.

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Aluminum

- Similar to copper, aluminum is utilized in its pure form because of its excellent thermal and electrical conductivity, ease of workability, and resistance to corrosion.
- Pure aluminum, which is soft and has a silvery-white appearance, belongs to the boron group of metals and has an atomic number of 13.
- In addition to its various alloys, pure aluminum is commonly used in power lines, beer kegs, window frames, automobiles, and kitchen utensils.
- When exposed to oxygen, aluminum develops a protective oxide layer that safeguards its surface from environmental damage.

Iron

- Iron is a hard, brittle metal that corrodes when exposed to moisture and high temperatures.
- Most of the iron mined is used to produce steel by alloying it with carbon. Its widespread use is due to its affordability and strength.
- Iron is a gray, silvery metal with magnetic properties and reacts readily with acids.
- It rusts easily in the presence of air and water, which is why it is typically alloyed to maintain its strength.
- Pure iron is 99.8% iron, with small amounts of carbon, manganese, and other elements.



Nickel

- ✚ Nickel, similar to aluminum, develops a passivation layer when exposed to oxygen, which safeguards it against corrosion.
- ✚ While both nickel and aluminum are silver-white metals, nickel has a distinct golden hue.
- ✚ Nickel is a hard, ductile metal with chemical reactivity and is magnetic at room temperature.
- ✚ Its corrosion resistance makes it a popular choice for alloys and coatings. Nickel is commonly used in products such as wiring, batteries, and electrodes.
- ✚ Additionally, like copper and aluminum, nickel exhibits excellent thermal conductivity, which is why it is also utilized in heat exchangers.

Chromium

- ✚ Chromium is a pure metal known for its ability to form a protective oxide layer. It is a lustrous, brittle, and hard metal primarily used in producing stainless steel and for plating, thanks to its bright, polished finish.
- ✚ Chromium can exist in various oxidation states ranging from -2 to +6, with the +0, +3, and +6 states being the most stable.
- ✚ Its widespread use stems from its shiny appearance, which enhances the aesthetic appeal of products like cars and household appliances.

Zinc

- ✚ Pure zinc is utilized as a protective coating because of its excellent corrosion resistance. This bluish-white metal is commonly used in alloys.
- ✚ In the galvanization process, a layer of zinc is applied to iron and steel, significantly enhancing their resistance to corrosion.
- ✚ Zinc is also alloyed with aluminum and copper to boost their strength, durability, and other properties.
- ✚ As the zinc content increases, copper and aluminum alloys become stronger, more durable, and highly resilient, though they also become more challenging to work with.

Titanium

- ✚ Titanium boasts a high strength-to-weight ratio, is highly resistant to rust, and can withstand a wide range of chemicals.

✚ The pure titanium grades are 1, 2, 3, and 4. Grade 1 is the softest and most formable, while Grade 2 offers increased strength. Grade 3 provides even more strength, is weldable, and highly resistant to corrosion.

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✚ Grade 4 is the strongest of the pure titanium grades, though it is the least malleable. It is durable, strong, and weldable, and can be cold worked. Among pure metals, titanium is particularly versatile, as it can be used in its pure form or alloyed with other metals to create exceptionally durable materials.

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Different types of metal alloys

The different types of metal alloys are:

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